

# 4 Interference of Light Waves

In previous Chapter, we studied light rays passing through a lens or reflecting from a mirror to describe the formation of images. This discussion completed our study of geometric optics. In this chapter and in next Chapter, we are concerned with wave optics or physical optics, the study of interference, diffraction, and polarization of light. These phenomena cannot be adequately explained with the ray optics used in next Chapters. We now learn how treating light as waves rather than as rays leads to a satisfying description of such phenomena.

## 4.1 Conditions for Interference

In Chapter 18, we studied the waves in interference model and found that the superposition of two mechanical waves can be constructive or destructive. In constructive interference, the amplitude of the resultant wave is greater than that of either individual wave, whereas in destructive interference, the resultant amplitude is less than that of the larger wave. Light waves also interfere with one another. Fundamentally, all interference associated with light waves arises when the electromagnetic fields that constitute the individual waves combine. If two lightbulbs are placed side by side so that light from both bulbs combines, no interference effects are observed because the light waves from one bulb are emitted independently of those from the other bulb. The emissions from the two lightbulbs do not maintain a constant phase relationship with each other over time. Light waves from an ordinary source such as a lightbulb undergo random phase changes in time intervals of less than a nanosecond. Therefore, the conditions for constructive interference, destructive interference, or some intermediate state are maintained only for such short time intervals. Because the eye cannot follow such rapid changes, no interference effects are observed. Such light sources are said to be incoherent. To observe interference of waves from two sources, the following conditions must be met:

- The sources must be coherent; that is, they must maintain a constant phase with respect to each other.
- The sources should be monochromatic; that is, they should be of a single wavelength.

As an example, single-frequency sound waves emitted by two side-by-side loudspeakers driven by a single amplifier can interfere with each other because the two speakers are coherent. In other words, they respond to the amplifier in the same way at the same time.

## 4.2 Young's Double-Slit Experiment

A common method for producing two coherent light sources is to use a monochromatic source to illuminate a barrier containing two small openings, usually in the shape of slits. The light emerging from the two slits is coherent because a single source produces the original light beam and the two slits serve only to separate the original beam into two parts (which, after all, is what is done to the sound signal from two side-by-side loudspeakers). Any random change in the light emitted by the source occurs in both beams at the same time. As a result, interference effects can be observed when the light from the two slits arrives at a viewing screen.

If the light traveled only in its original direction after passing through the slits as shown in Figure 4.1a, the waves would not overlap and no interference pattern would be seen. Instead, as we have discussed in our treatment of Huygens's principle (Section 35.6), the waves spread out from the slits as shown in Figure 4.1b. In other words, the light deviates from a straight-line path and enters the region that would otherwise be shadowed. As noted in Section 35.3, this divergence of light from its initial line of travel is called diffraction.

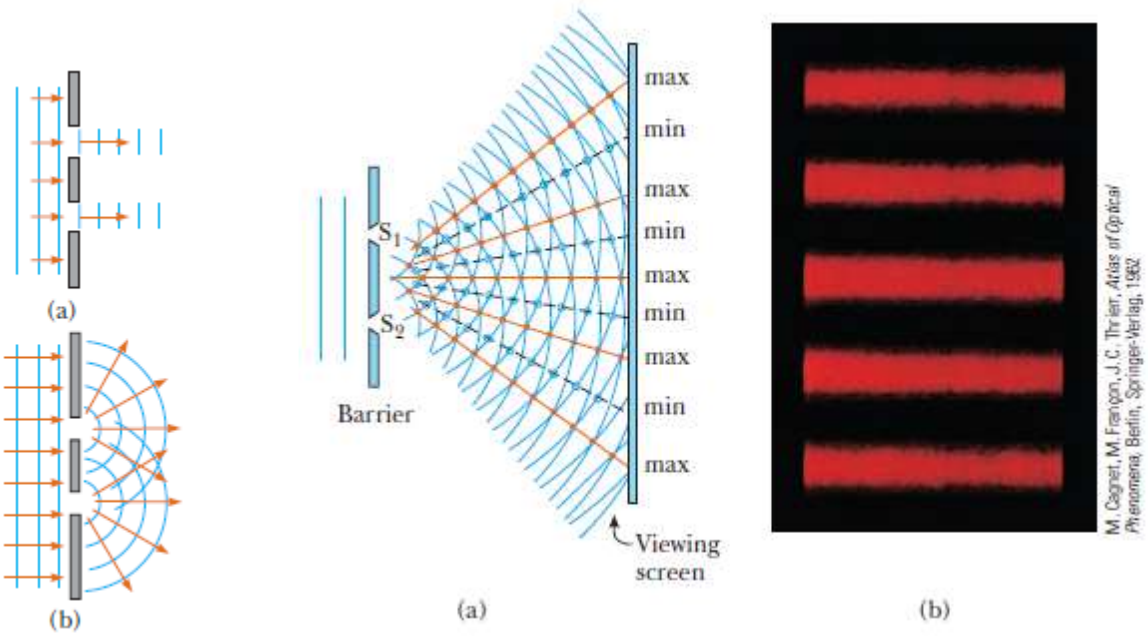



Figure 4.1 ACTIVE FIGURE 4.2

Interference in light waves from two sources was first demonstrated by Thomas Young in 1801. A schematic diagram of the apparatus Young used is shown in Active Figure 4.2a. Plane light waves arrive

at a barrier that contains two parallel slits S1 and S2. The light from S1 and S2 produces on a viewing screen a visible pattern of bright and dark parallel bands called fringes (Active Fig. 4.2b). When the light from S1 and that from S2 both arrive at a point on the screen such that constructive interference occurs at that location, a bright fringe appears. When the light from the two slits combines destructively at any location on the screen, a dark fringe results. Figure 4.3 is a photograph of an interference pattern produced by two coherent vibrating sources in a water tank. Figure 4.4 shows some of the ways in which two waves can combine at the screen. In Figure 4.4a, the two waves, which leave the two slits in phase, strike the screen at the central point O. Because both waves travel the same distance, they arrive at O in phase. As a result, constructive interference occurs at this location and a bright fringe is observed. In Figure 4.4b, the two waves also start in phase, but here the lower wave has to travel one wavelength farther than the upper wave to reach point P. Because the lower wave falls behind the upper one by exactly one wavelength, they still arrive in phase at P and a second bright fringe appears at this location. At point R in Figure 4.4c, however, between points O and P, the lower wave has fallen half a wavelength behind the upper wave and a trough of the upper wave overlaps a crest of the lower wave, giving rise to destructive interference at point R. A dark fringe is therefore observed at this location.

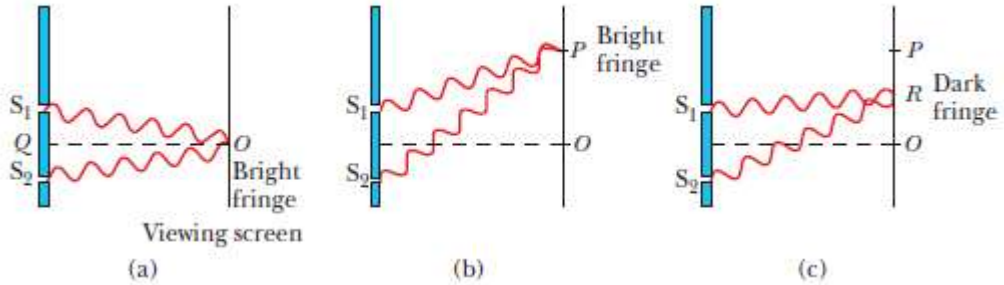



Figure 4.4 (a) Constructive interference occurs at point O when the waves combine. (b) Constructive interference also occurs at point P. (c) Destructive interference occurs at point R when the two waves combine because the lower wave falls one-half a wavelength behind the upper wave. (All figures not to scale.)

## 4.3 Light Waves in Interference

We can describe Young's experiment quantitatively with the help of Figure 4.5. The viewing screen is located a perpendicular distance L from the barrier containing two slits, S1 and S2 (Fig. 4.5a). These slits are separated by a distance d, and the source is monochromatic. To reach any arbitrary point P in the upper half of the screen, a wave from the lower slit must travel farther than a wave from the upper slit by a distance d sinθ (Fig. 4.5b). This distance is called the path difference δ (Greek letter delta). If we assume the rays labeled r1 and r2 are parallel, which is approximately true if L is much greater than d, then δ is given by

$$\delta = r_2 - r_1 = d \sin\theta \qquad (4.1)$$

The value of δ determines whether the two waves are in phase when they arrive at point P. If δ is either zero or some integer multiple of the wavelength, the two waves are in phase at point P and constructive interference results. Therefore, the condition for bright fringes, or **constructive interference**, at point P is

$$d \sin \theta_{\text{bright}} = m\lambda \quad (m = 0, \pm 1, \pm 2, \cdots)$$

**(4.2)**

The number m is called the **order number**. For constructive interference, the order number is the same as the number of wavelengths that represents the path difference between the waves from the two slits. The central bright fringe at $\theta_{\text{bright}} = 0$ is called the ***zeroth-order maximum***. The first maximum on either side, where m = ±1, is called the ***first-order maximum***, and so forth.

When $\delta$ is an odd multiple of $\lambda/2$, the two waves arriving at point P are 180° out of phase and give rise to destructive interference. Therefore, the condition for dark fringes, or ***destructive interference***, at point P is

$$d \sin \theta_{\text{dark}} = (m + \tfrac{1}{2})\lambda \quad (m = 0, \pm 1, \pm 2, \cdots)$$

**(4.3)**

These equations provide the angular positions of the fringes. It is also useful to obtain expressions for the linear positions measured along the screen from O to P. From the triangle OPQ in Figure 37.5a, we see that

$$\tan \theta = \frac{y}{L}$$

**(4.4)**

Using this result, the linear positions of bright and dark fringes are given by

$$y_{\text{bright}} = L \tan \theta_{\text{bright}}$$
$$y_{\text{dark}} = L \tan \theta_{\text{dark}}$$

**(4.5)**

where $\theta_{\text{bright}}$ and $\theta_{\text{dark}}$ are given by Equations 4.2 and 4.3.

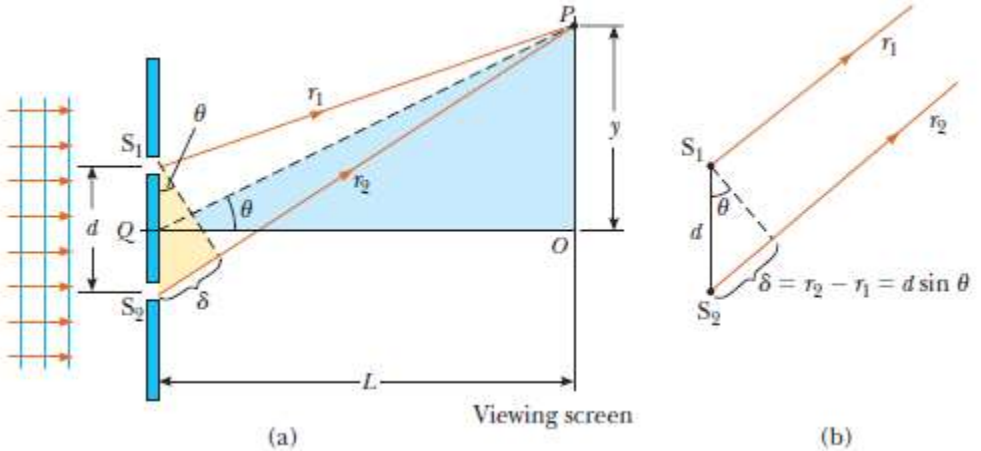



Figure 4.5

When the angles to the fringes are small, the positions of the fringes are linear near the center of the pattern. This can be verified by noting that for small angles, $\tan\theta \approx \sin\theta$, so Equation 4.5 gives the positions of the bright fringes as $y_{\text{bright}} = L \sin \theta_{\text{bright}}$. Incorporating Equation 4.2 gives

$$y_{\text{bright}} = L\left(\frac{m\lambda}{d}\right) \quad \text{(small angles)}$$

**(4.6)**

This result shows that $y_{bright}$ is linear in the order number m, so the fringes are equally spaced. Young's double-slit experiment provides a method for measuring the wavelength of light. In fact, Young used this technique to do precisely that. In addition, his experiment gave the wave model of light a great deal of credibility. It was inconceivable that particles of light coming through the slits could cancel one another in a way that would explain the dark fringes. The principles discussed in this section are the basis of the waves in interference analysis model. This model was applied to mechanical waves in one dimension in Chapter 18. Here we see the details of applying this model in three dimensions to light.

EXAMPLE 1. **Measuring the Wavelength of a Light Source**

A viewing screen is separated from a double slit by 1.2 m. The distance between the two slits is 0.030 mm. Monochromatic light is directed toward the double slit and forms an interference pattern on the screen. The second-order bright fringe (m = 2) is 4.5 cm from the center line on the screen. (A) Determine the wavelength of the light.

**SOLUTION**

**Conceptualize** Study Figure 37.5 to be sure you understand the phenomenon of interference of light waves.

**Categorize** We evaluate results using equations developed in this section, so we categorize this example as a substitution problem.

Solve Equation 37.7 for the wavelength and substitute numerical values:

$$\lambda = \frac{y_{bright} d}{mL} = \frac{(4.5 \times 10^{-2}\ \text{m})(3.0 \times 10^{-5}\ \text{m})}{2(1.2\ \text{m})}$$

$$= 5.6 \times 10^{-7}\ \text{m} = \boxed{560\ \text{nm}}$$

**(B)** Calculate the distance between adjacent bright fringes.

**SOLUTION**

Find the distance between adjacent bright fringes from Equation 37.7 and the results of part (A):

$$y_{m+1} - y_m = L\frac{(m+1)\lambda}{d} - L\left(\frac{m\lambda}{d}\right)$$

$$= L\left(\frac{\lambda}{d}\right) = 1.2\ \text{m}\left(\frac{5.6 \times 10^{-7}\ \text{m}}{3.0 \times 10^{-5}\ \text{m}}\right)$$

$$= 2.2 \times 10^{-2}\ \text{m} = \boxed{2.2\ \text{cm}}$$

EXAMPLE 2. **Separating Double-Slit Fringes of Two Wavelengths**

A light source emits visible light of two wavelengths: $\lambda$ = 430 nm and $\lambda'$= 510 nm. The source is used in a double slit interference experiment in which L = 1.50 m and d = 0.025 0 mm. Find the separation distance between the third-order bright fringes for the two wavelengths.

**SOLUTION**

**Conceptualize** In Figure 37.5a, imagine light of two wavelengths incident on the slits and forming two interference patterns on the screen.

**Categorize** We evaluate results using equations developed in this section, so we categorize this example as a substitution problem.

Use Equation 37.7, with $m = 3$, to find the fringe positions corresponding to these two wavelengths:

$$y_{bright} = L\left(\frac{m\lambda}{d}\right) = L\left(\frac{3\lambda}{d}\right) = 1.50\ \text{m}\left[\frac{3(430 \times 10^{-9}\ \text{m})}{0.025\,0 \times 10^{-3}\ \text{m}}\right]$$

$$= 7.74 \times 10^{-2}\ \text{m}$$

$$y'_{bright} = L\left(\frac{m\lambda'}{d}\right) = L\left(\frac{3\lambda'}{d}\right) = 1.50\ \text{m}\left[\frac{3(510 \times 10^{-9}\ \text{m})}{0.025\,0 \times 10^{-3}\ \text{m}}\right]$$

$$= 9.18 \times 10^{-2}\ \text{m}$$

Evaluate the separation distance between the two fringes:

$$\Delta y = 9.18 \times 10^{-2}\ \text{m} - 7.74 \times 10^{-2}\ \text{m}$$

$$= 1.44 \times 10^{-2}\ \text{m} = \boxed{1.44\ \text{cm}}$$

**What If?** What if we examine the entire interference pattern due to the two wavelengths and look for overlapping fringes? Are there any locations on the screen where the bright fringes from the two wavelengths overlap exactly?

**Answer** Find such a location by setting the location of any bright fringe due to $\lambda$ equal to one due to $\lambda'$, using Equation 37.7:

$$L\left(\frac{m\lambda}{d}\right) = L\left(\frac{m'\lambda'}{d}\right) \rightarrow \frac{m'}{m} = \frac{\lambda}{\lambda'}$$

Substitute the wavelengths:

$$\frac{m'}{m} = \frac{430\text{ nm}}{510\text{ nm}} = \frac{43}{51}$$

Use Equation 37.7 to find the value of $y$ for these fringes:

$$y = 1.50\text{ m}\left[\frac{51(430 \times 10^{-9}\text{ m})}{0.025\,0 \times 10^{-3}\text{ m}}\right] = 1.32\text{ m}$$

This value of $y$ is comparable to $L$, so the small-angle approximation used for Equation 37.7 is *not* valid. This conclusion suggests we should not expect Equation 37.7 to give us the correct result. If you use Equation 37.5, you can show that the bright fringes do indeed overlap when the same condition, $m'/m = \lambda/\lambda'$, is met (see Problem 38). Therefore, the 51st fringe of the 430-nm light does overlap with the 43rd fringe of the 510-nm light, but not at the location of 1.32 m. You are asked to find the correct location as part of Problem 38.

## 4.4 Intensity Distribution of the Double-Slit Interference Pattern

Notice that the edges of the bright fringes in Active Figure 4.2b are not sharp; rather, there is a gradual change from bright to dark. So far, we have discussed the locations of only the centers of the bright and dark fringes on a distant screen. Let's now direct our attention to the intensity of the light at other points between the positions of maximum constructive and destructive interference. In other words, we now calculate the distribution of light intensity associated with the double slit interference pattern.

Again, suppose the two slits represent coherent sources of sinusoidal waves such that the two waves from the slits have the same angular frequency $\omega$ and are in phase. The total magnitude of the electric field at point P on the screen in Figure 4.6 is the superposition of the two waves. Assuming that the two waves have the same amplitude $E_0$, we can write the magnitude of the electric field at point P due to each wave separately as

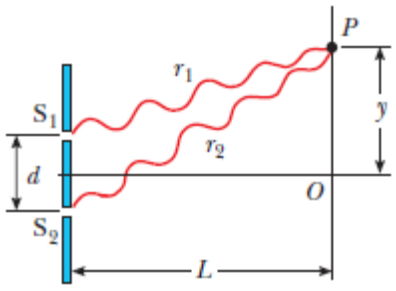



Figure 4.6

$$E_1 = E_0 \sin \omega t \quad \text{and} \quad E_2 = E_0 \sin(\omega t + \phi) \tag{4.7}$$

Although the waves are in phase at the slits, their phase difference $\phi$ at P depends on the path difference $\delta = r_2 - r_1 = d \sin\theta$. A path difference of $\lambda$ (for constructive interference) corresponds to a phase difference of $2\pi$ rad. A path difference of $\delta$ is the same fraction of $\lambda$ as the phase difference $\phi$ is of $2\pi$. We can describe this fraction mathematically with the ratio

$$\frac{\delta}{\lambda} = \frac{\phi}{2\pi}$$

which gives

$$\phi = \frac{2\pi}{\lambda}\delta = \frac{2\pi}{\lambda} d \sin\theta \tag{4.8}$$

This equation shows how the phase difference $\phi$ depends on the angle $\theta$ in Figure 4.5. Using the superposition principle and Equation 4.8, we obtain the following expression for the magnitude of the resultant electric field at point P :

$$E_P = E_1 + E_2 = E_0[\sin \omega t + \sin (\omega t + \phi)] \tag{4.9}$$

We can simplify this expression by using the trigonometric identity

$$\sin A + \sin B = 2 \sin \left(\frac{A + B}{2}\right) \cos \left(\frac{A - B}{2}\right)$$

Taking A = $\omega$t + $\phi$ and B = $\omega$t, Equation 4.9 becomes

$$E_P = 2E_0 \cos \left(\frac{\phi}{2}\right) \sin \left(\omega t + \frac{\phi}{2}\right) \tag{4.10}$$

This result indicates that the electric field at point P has the same frequency $\omega$ as the light at the slits but that the amplitude of the field is multiplied by the factor 2 cos ($\phi$/2). To check the consistency of this result, note that if $\phi$ = 0, 2$\pi$, 4$\pi$, . . . , the magnitude of the electric field at point P is $2E_0$, corresponding to the condition for maximum constructive interference. These values of $\phi$ are consistent with Equation 4.2 for constructive interference. Likewise, if $\phi$ = $\pi$, 3$\pi$, 5$\pi$, . . . , the magnitude of the electric field at point P is zero, which is consistent with Equation 4.3 for total destructive interference.

Finally, to obtain an expression for the light intensity at point P, recall from Section 4.4 that the intensity of a wave is proportional to the square of the resultant electric field magnitude at that point. Using Equation 4.10, we can therefore express the light intensity at point P as

$$I \propto E_P^2 = 4E_0^2 \cos^2 \left(\frac{\phi}{2}\right) \sin^2 \left(\omega t + \frac{\phi}{2}\right)$$

Most light-detecting instruments measure time-averaged light intensity, and the time-averaged value of $\sin^2 (\omega t + \phi/2)$ over one cycle is 1/2. Therefore, we can write the average light intensity at point P as

$$I = I_{\max} \cos^2 \left(\frac{\phi}{2}\right) \tag{4.11}$$

where $I_{max}$ is the maximum intensity on the screen and the expression represents the time average. Substituting the value for $\phi$ given by Equation 4.9 into this expression gives

$$I = I_{\max} \cos^2 \left(\frac{\pi d \sin \theta}{\lambda}\right) \tag{4.12}$$

Alternatively, because $\sin \theta \approx y/L$ for small values of $\theta$ in Figure 4.5, we can write Equation 4.12 in the form

$$I = I_{\max} \cos^2 \left(\frac{\pi d}{\lambda L} y\right) \tag{4.13}$$

Constructive interference, which produces light intensity maxima, occurs when the quantity $\pi$ dy/$\lambda$L is an integral multiple of $\pi$, corresponding to y = ($\lambda$L/d)m. This result is consistent with Equation 4.7.

A plot of light intensity versus dsin$\theta$ is given in Figure 4.7. The interference pattern consists of equally spaced fringes of equal intensity. Remember, however, that this result is valid only if the slit-to-screen distance L is much greater than the slit separation and only for small values of $\theta$.

Figure 4.8 shows similar plots of light intensity versus dsin$\theta$ for light passing through multiple slits. For more than two slits, the pattern contains primary and secondary maxima. For three slits, notice that the primary maxima are nine times more intense than the secondary maxima as measured by the height of the curve because the intensity varies as $E^2$. For N slits, the intensity of the primary maxima is $N^2$ times greater than that due to a single slit. As the number of slits increases, the primary maxima increase in intensity and become narrower, while the secondary maxima decrease in intensity relative to the primary maxima. Figure 4.8 also shows that as the number of slits increases, the number of secondary maxima also increases. In fact, the number of secondary maxima is always N-2, where N is the number of slits. In Section 5.4, we shall investigate the pattern for a very large number of slits in a device called a diffraction grating.

## 4.5 Diffraction

Suppose a light beam is incident on two slits, as in Young's double-slit experiment. If the light truly traveled in straight-line paths after passing through the slits, as in Figure 4.7a, the waves wouldn't overlap and no interference pattern would be seen. Instead, Huygens' principle requires that the waves spread out from the slits, as shown in Figure 4.7b. In other words, the light bends from a straight-line path and enters the region that would otherwise be shadowed. This spreading out of light from its initial line of travel is called diffraction.

In general, diffraction occurs when waves pass through small openings, around obstacles, or by sharp edges. For example, when a single narrow slit is placed between a distant light source (or a laser beam) and a screen, the light produces a diffraction pattern like that in Figure 4.8. The pattern consists of a broad, intense central band flanked by a series of narrower, less intense secondary bands (called secondary maxima) and a series of dark bands, or minima. This phenomenon can't be explained within the framework of geometric optics, which says that light rays traveling in straight lines should cast a sharp image of the slit on the screen.

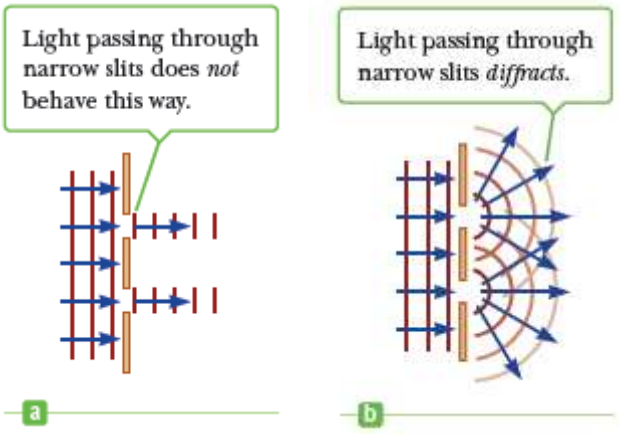



**Figure 4.7**

Figure 4.9 shows the diffraction pattern and shadow of a penny. The pattern consists of the shadow, a bright spot at its center, and a series of bright and dark circular bands of light near the edge of the shadow. The bright spot at the center (called the Fresnel bright spot) is explained by **Augustin Fresnel**'s wave theory of light, which predicts constructive interference at this point for certain locations of the penny. From the viewpoint of geometric optics, there shouldn't be any bright

spot: the center of the pattern would be completely screened by the penny. One type of diffraction, called **Fraunhofer diffraction**, occurs when the rays leave the diffracting object in parallel directions. Fraunhofer diffraction can be achieved experimentally either by placing the observing screen far from the slit or by using a converging lens to focus the parallel rays on a nearby screen, as in
Active Figure 4.10a. A bright fringe is observed along the axis at $\theta = 0$, with alternating dark and bright fringes on each side of the central bright fringe. Active Figure 4.10b is a photograph of a single-slit Fraunhofer diffraction pattern.

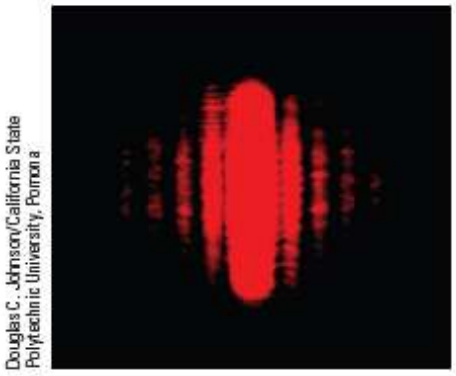



**Figure 4.8**

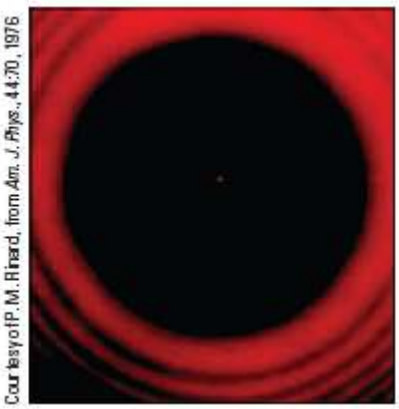
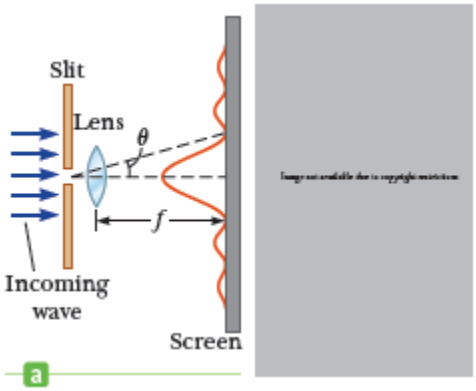



**Figure 4.9**




**Active Figure 4.10**

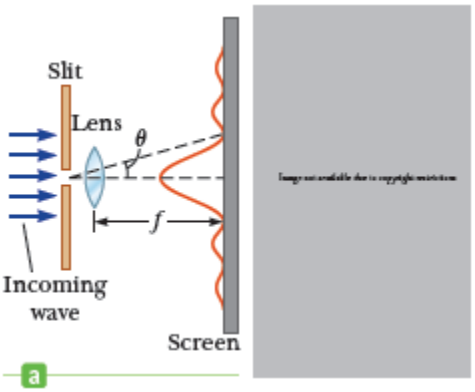
## 4.6 Signle-Slit Diffraction

Until now we have assumed slits have negligible width, acting as line sources of light. In this section we determine how their nonzero widths are the basis for understanding the nature of the Fraunhofer diffraction pattern produced by a single slit.

We can deduce some important features of this problem by examining waves coming from various portions of the slit, as shown in Figure 4.11. According to Huygens' principle, each portion of the slit acts as a source of waves. Hence, light from one portion of the slit can interfere with light from another portion, and the resultant intensity on the screen depends on the direction $\theta$.

To analyze the diffraction pattern, it's convenient to divide the slit into halves, as in Figure 4.12. All the waves that originate at the slit are in phase. Consider waves 1 and 3, which originate at the bottom and center of the slit, respectively. Wave 1 travels farther than wave 3 by an amount equal

to the path difference (a/2) sinθ, where a is the width of the slit. Similarly, the path difference between waves 3 and 5 is (a/2) sinθ. If this path difference is exactly half of a wavelength (corresponding to a phase difference of 180°), the two waves cancel each other, and destructive interference results. This is true, in fact, for any two waves that originate at points separated by half the slit width because the phase difference between two such points is 180°. Therefore, waves from the upper half of the slit interfere destructively with waves from the lower half of the slit when




**Active Figure 4.11**

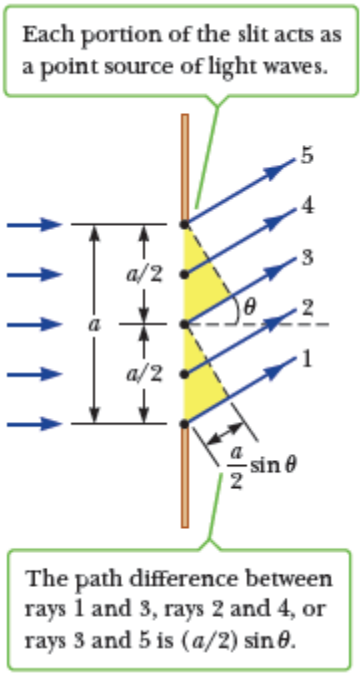



**Figure 4.12**

$$\frac{a}{2}\sin\theta = \frac{\lambda}{2} \quad \text{or when} \quad \sin\theta = \frac{\lambda}{a}$$

If we divide the slit into four parts rather than two and use similar reasoning, we find that the screen is also dark when

$$\sin\theta = \frac{2\lambda}{a}$$

Continuing in this way, we can divide the slit into six parts and show that darkness occurs on the screen when

$$\sin\theta = \frac{3\lambda}{a}$$

Therefore, the general condition for **destructive interference** for a single slit of width a is

$$\sin\theta_{\text{dark}} = m\frac{\lambda}{a} \qquad m = \pm 1, \pm 2, \pm 3, \ldots \tag{4.14}$$

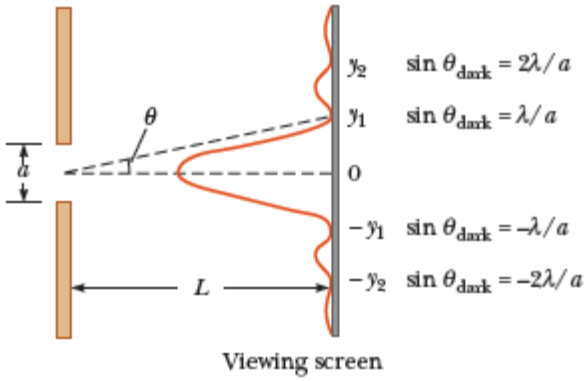



**Figure 4.13**

Equation 4.14 gives the values of θ for which the diffraction pattern has zero intensity, where a dark fringe forms. The equation tells us nothing about the variation in intensity along the screen, however. The general features of the intensity distribution along the screen are shown in Figure 4.13. A broad central bright fringe is flanked by much weaker bright fringes alternating with dark fringes. The various dark fringes (points of zero intensity) occur at the values of θ that satisfy Equation 4.14. The points of constructive interference lie approximately halfway between the dark fringes. Note that the central bright fringe is twice as wide as the weaker maxima having m > 1.

PROBLEM Light of wavelength $5.80\times10^2$ nm is incident on a slit of width 0.300 mm. The observing screen is placed 2.00 m from the slit. Find the positions of the first dark fringes and the width of the central bright fringe.

STRATEGY This problem requires substitution into Equation 4.14 to find the sines of the angles of the first dark fringes. The positions can then be found with the tangent function because for small angles $\sin\theta \approx \tan\theta$. The extent of the central maximum is defined by these two dark fringes.

**SOLUTION**

The first dark fringes that flank the central bright fringe correspond to $m = \pm1$ in Equation 24.11:

$$\sin\theta = \pm\frac{\lambda}{a} = \pm\frac{5.80\times10^{-7}\text{ m}}{0.300\times10^{-3}\text{ m}} = \pm1.93\times10^{-3}$$

Use the triangle in Figure 24.18 to relate the position of the fringe to the tangent function:

$$\tan\theta = \frac{y_1}{L}$$

Because $\theta$ is very small, we can use the approximation $\sin\theta \approx \tan\theta$ and then solve for $y_1$:

$$\sin\theta \approx \tan\theta \approx \frac{y_1}{L}$$

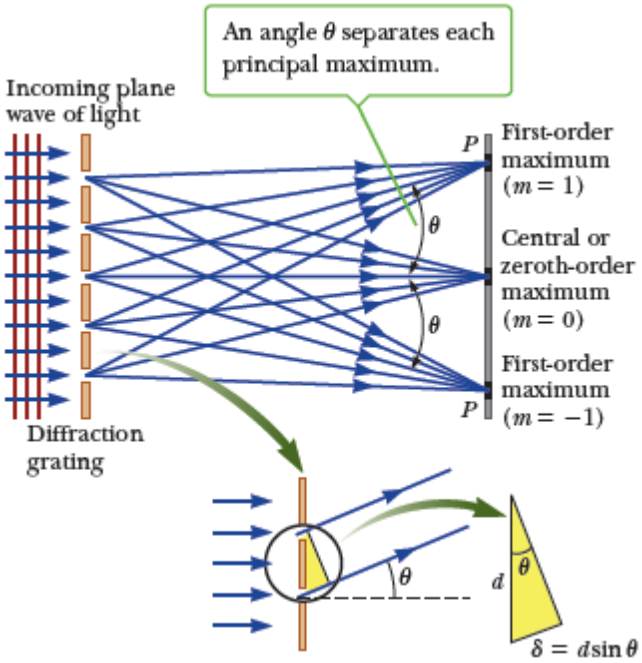
$$y_1 \approx L\sin\theta = (2.00\text{ m})(\pm1.93\times10^{-3}) = \pm\,3.86\times10^{-3}\text{ m}$$

Compute the distance between the positive and negative first-order maxima, which is the width $w$ of the central maximum:

$$w = +3.86\times10^{-3}\text{ m} - (-3.86\times10^{-3}\text{ m}) = 7.72\times10^{-3}\text{ m}$$

## 4.7 The Diffraction Grating

The diffraction grating, a useful device for analyzing light sources, consists of a large number of equally spaced parallel slits. A grating can be made by scratching parallel lines on a glass plate with a precision machining technique. The clear panes between scratches act like slits. A typical grating contains several thousand lines per centimeter. For example, a grating ruled with 5 000 lines/cm has a slit spacing d equal to the reciprocal of that number; hence, d = (1/5 000) cm = $2\times10^{-4}$ cm. Active Figure 4.14 is a schematic diagram of a section of a plane diffraction grating. A plane wave is incident from the left, normal to the plane of the grating.



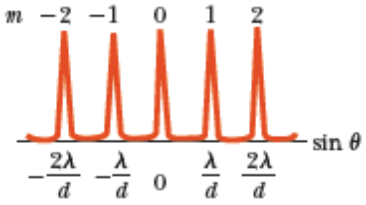

**Active Figure 4.14**

The intensity of the pattern on the screen is the result of the combined effects of interference and diffraction. Each slit causes diffraction, and the diffracted beams in turn interfere with one another to produce the pattern. Moreover, each slit acts as a source of waves, and all waves start in phase at the slits. For some arbitrary direction u measured from the horizontal, however, the waves must travel different path lengths before reaching a particular point P on the screen. In Figure 4.14, note that the path difference between waves from any two adjacent slits is dsinθ. If this path difference equals one wavelength or some integral multiple of a wavelength, waves from all slits will be in phase at P and a bright line will be observed
at that point. Therefore, the condition for maxima in the interference pattern at the angle θ is

$$d \sin \theta_{\text{bright}} = m\lambda \qquad m = 0, \pm 1, \pm 2, \ldots$$

**(4.15)**

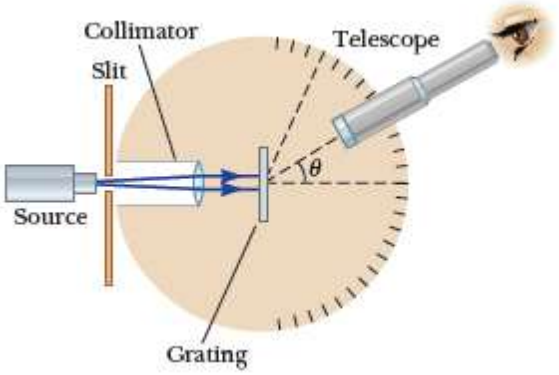
Light emerging from a slit at an angle other than that for a maximum interferes nearly completely destructively with light from some other slit on the grating. All such pairs will result in little or no transmission in that direction, as illustrated in Active Figure 4.15..




**Active Figure 4.15**

Equation 4.15 can be used to calculate the wavelength from the grating spacing and the angle of deviation, θ. The integer m is the order number of the diffraction pattern. If the incident radiation contains several wavelengths, each wavelength deviates through a specific angle, which can be found from Equation 4.15. All wavelengths are focused at θ = 0, corresponding to m = 0. This point is called the zeroth-order maximum. *The first-order maximum*, corresponding to m = 1, is observed at an angle that satisfies the relationship sinθ = λ/d; *the second-order maximum*, corresponding to m = 2, is observed at a larger angle θ, and so on. Active Figure 4.15 is a sketch of the intensity distribution for some of the orders produced by a diffraction grating. Note the sharpness of the principal maxima and the broad range of the dark areas, a pattern in direct contrast to the broad bright fringes characteristic of the two-slit interference pattern. A simple arrangement that can be used to measure the angles in a diffraction pattern is shown in Active Figure 4.16. This setup is a form of

a diffraction grating spectrometer. The light to be analyzed passes through a slit and is formed into a parallel beam by a lens. The light then strikes the grating at a 90° angle. The diffracted light leaves the grating at angles that satisfy Equation 4.15. A telescope is used to view the image of the slit. The wavelength can be determined by measuring the angles at which the images of the slit appear for the various orders.




**Active Figure 4.16**

PROBLEM Monochromatic light from a helium–neon laser ($\lambda$ = 632.8 nm) is incident normally on a diffraction grating containing $6.00 \times 10^3$ lines/cm. Find the angles at which one would observe the first-order maximum, the second-order maximum, and so forth.

STRATEGY Find the slit separation by inverting the number of lines per centimeter, then substitute values into Equation 4.15.

**SOLUTION**

Invert the number of lines per centimeter to obtain the slit separation:

$$d = \frac{1}{6.00 \times 10^3\,\text{cm}^{-1}} = 1.67 \times 10^{-4}\,\text{cm} = 1.67 \times 10^3\,\text{nm}$$

Substitute $m = 1$ into Equation 24.12 to find the sine of the angle corresponding to the first-order maximum:

$$\sin\theta_1 = \frac{\lambda}{d} = \frac{632.8\,\text{nm}}{1.67 \times 10^3\,\text{nm}} = 0.379$$

Take the inverse sine of the preceding result to find $\theta_1$:

$$\theta_1 = \sin^{-1} 0.379 = 22.3°$$

Repeat the calculation for $m = 2$:

$$\sin\theta_2 = \frac{2\lambda}{d} = \frac{2(632.8\,\text{nm})}{1.67 \times 10^3\,\text{nm}} = 0.758$$

$$\theta_2 = 49.3°$$

Repeat the calculation for $m = 3$:

$$\sin\theta_3 = \frac{3\lambda}{d} = \frac{3(632.8\,\text{nm})}{1.67 \times 10^3\,\text{nm}} = 1.14$$

Because $\sin\theta$ can't exceed 1, there is no solution for $\theta_3$.

## PROBLEMS

1. A laser beam ($\lambda$ = 632.8 nm) is incident on two slits 0.200 mm apart. How far apart are the bright interference fringes on a screen 5.00 m away from the double slits?
2. A Young's interference experiment is performed with monochromatic light. The separation between the slits is 0.500 mm, and the interference pattern on a screen 3.30 m away shows the first side maximum 3.40 mm from the center of the pattern. What is the wavelength?
3. Two radio antennas separated by 300 m as shown in Figure P4.3 simultaneously broadcast identical signals at the same wavelength. A radio in a car traveling due north receives the signals. (a) If the car is at the position of the second maximum, what is the wavelength of the signals? (b) How much farther must the car travel to encounter the next minimum in reception? Note: Do not use the small-angle approximation in this problem.

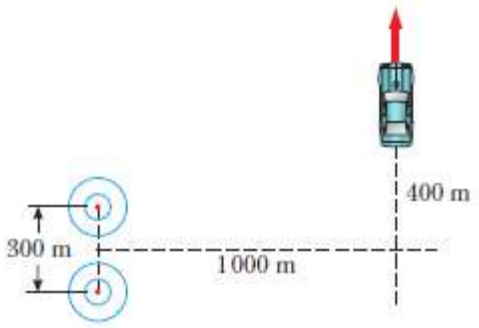


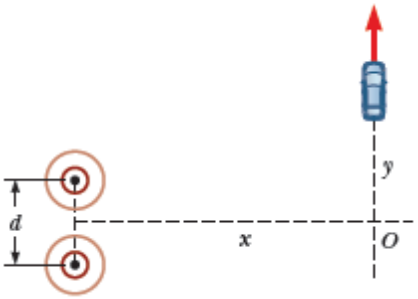

Figure P4.3

4. In a location where the speed of sound is 354 m/s, a 2 000-Hz sound wave impinges on two slits 30.0 cm apart. (a) At what angle is the first maximum located? (b) What If? If the sound wave is replaced by 3.00-cm microwaves what slit separation gives the same angle for the first maximum? (c) What If? If the slit separation is 1.00 mm, what frequency of light gives the same first maximum angle?
5. Young's double-slit experiment is performed with 589-nm light and a distance of 2.00 m between the slits and the screen. The tenth interference minimum is observed 7.26 mm from the central maximum. Determine the spacing of the slits.

6. A laser beam is incident on two slits with a separation of 0.200 mm, and a screen is placed 5.00 m from the slits. If the bright interference fringes on the screen are separated by 1.58 cm, what is the wavelength of the laser light?
7. In a Young's double-slit experiment, a set of parallel slits with a separation of 0.100 mm is illuminated by light having a wavelength of 589 nm, and the interference pattern is observed on a screen 4.00 m from the slits. (a) What is the difference in path lengths from each of the slits to the location of a third-order bright fringe on the screen? (b) What is the difference in path lengths from the two slits to the location of the third dark fringe on the screen, away from the center of the pattern?
8. A pair of narrow, parallel slits separated by 0.250 mm is illuminated by the green component from a mercury vapor lamp ($\lambda$ = 546.1 nm). The interference pattern is observed on a screen 1.20 m from the plane of the parallel slits. Calculate the distance (a) from the central maximum to the first bright region on either side of the central maximum and (b) between the first and second dark bands in the interference pattern.
9. Light of wavelength 620 nm falls on a double slit, and the first bright fringe of the interference pattern is seen at an angle of 15.0° from the central maximum. Find the separation between the slits.

10. In a location where the speed of sound is 354 m/s, a 2 000-Hz sound wave impinges on two slits 30.0 cm apart. (a) At what angle is the first maximum located? (b) If the sound wave is replaced by 3.00-cm microwaves, what slit separation gives the same angle for the first maximum? (c) If the slit separation is 1.00 mm, what frequency of light gives the same first maximum angle?
11. A double slit separated by 0.058 0 mm is placed 1.50 m from a screen. (a) If yellow light of wavelength 588 nm strikes the double slit, what is the separation between the zeroth-order and first-order maxima on the screen? (b) If blue light of wavelength 412 nm strikes the double slit, what is the separation between the second-order and fourth-order maxima?

12. Two radio antennas separated by d 5 300 m, as shown in following Figure, simultaneously broadcast identical signals at the same wavelength. A car travels due north along a straight line at position x = 1 000 m from the center point between the antennas, and its radio receives the signals. (a) If the car is at the position of the second maximum after that at point O when it has traveled a distance of y = 400 m northward, what is the wavelength of the signals? (b) How much farther must the car travel from this position to encounter the next minimum in reception? Hint: Do not use the small angle approximation in this problem.




13. Light of wavelength 6.0 × $10^2$ nm falls on a double slit, and the first bright fringe of the interference pattern is observed to make an angle of 12° with the horizontal. Find the separation between the slits.
14. Monochromatic light falls on a screen 1.75 m from two slits separated by 2.10 mm. The first- and second-order bright fringes are separated by 0.552 mm. What is the wavelength of the light?
15. A pair of slits, separated by 0.150 mm, is illuminated by light having a wavelength of $\lambda$ = 643 nm. An interference pattern is observed on a screen 140 cm from the slits. Consider a point on the screen located at y = 1.80 cm from the central maximum of this pattern. (a) What is the path difference d for the two slits at the location y? (b) Express this path difference in terms of the wavelength. (c) Will the interference correspond to a maximum, a minimum, or an intermediate condition?